# A Real USD SOFR Interest Rate Swap

## John Hancock Financial Opportunities Fund | Evidence, inferred timing & modelling assumptions

Purpose: a shareable specification for constructing an IRS valuation, yield-curve calibration, VaR/ES simulation and out-of-sample backtest. All unverified terms are explicitly marked; nothing in this brief constitutes the original trade confirmation.

## 1 Evidence status — how to read this brief

| VERIFIED [V] | MARKET EVIDENCE [M] | ASSUMED [A] / UNKNOWN [U] |
|---|---|---|
| Shown in John Hancock's published SEC fund holdings or financial-statement notes. | A separate, contemporaneous indicative market quote; not a John Hancock trade confirmation. | A project design choice [A], or a detail not established by the cited public documents [U]. |

## 2 Actual swap: verified public terms [V]

| Field | SEC disclosure / interpretation | Evidence |
|---|---|---|
| Fund / reporting entity | John Hancock Financial Opportunities Fund (BTO) | [S1], [S2] |
| Derivative / currency | Interest rate swap; USD | [S1], [S2] |
| Notional outstanding | USD 10,000,000 | [S1], [S2] |
| Fund pays | Fixed rate of 3.356% (annual rate) | [S1], [S2] |
| Fund receives | USD Compounded SOFR (also reported as SOFR Compounded OIS) | [S1], [S2] |
| Fixed / floating frequency | Semi-annual / quarterly | [S1], [S2] |
| Maturity as reported | May 2028 (month only) | [S1], [S2] |
| Clearing | Centrally cleared (original executing counterparty not named) | [S1] |
| Unamortized upfront payment | Reported as dash (—); NOT proof that no upfront payment ever existed | [S1] |
| Fund value on 30 Jun 2023 | USD +275,600; reported value/unrealized appreciation | [S2] |
| Fund value on 30 Jun 2026 | USD +117,354; reported value/unrealized appreciation | [S1] |

Important: the published "payments made / received" are the two contractual legs, not dollar amounts already transferred. The notional is a calculation base and ordinarily is not exchanged in a plain-vanilla single-currency IRS.

## 3 Can we infer when it was entered?

Best-supported working inference: around May 2023, not a verified execution date. The swap was in the fund's 30 June 2023 holdings [S2]. A separate May 2023 professional interest-rate-hedging presentation displays an indicative 5-year SOFR swap rate of 3.350% as of 17 May 2023, with May 2028 maturity [S3]. That is only 0.006 percentage points (0.6 basis points) from the fund's 3.356% fixed coupon.

This is compelling circumstantial consistency in both rate level and maturity month, but the [S3] rate is illustrative, not an executed quote, and different payment conventions, fees, mark-ups or forward starts can change fixed coupons. The coupon alone cannot uniquely recover the trade date. A five-year original tenor is also inferred, not disclosed.

## 4 Proposed model contract — NOT the original confirmation [A]

| Modelling item | Baseline assumption | Status |
|---|---|---|
| Illustrative trade date | 17 May 2023 | [A], motivated by [M] |
| Effective date | 19 May 2023 (T+2 business-day working assumption) | [A] |
| Exact maturity | 19 May 2028; modelled original 5-year tenor | [A] |
| Notional path | USD 10m constant, no amortization or notional exchange | [A] |
| Floating spread | 0 bp over compounded SOFR; no spread shown in this row | [A] |
| Fixed / floating day count | ACT/360 / ACT/360 | [A] |
| Calendar and adjustment | US Government Securities; Modified Following | [A] |
| Floating convention | Daily compounded SOFR in arrears; 2 business-day lookback | [A] |
| Payment lag | 2 business days after adjusted accrual period end | [A] |
| Discount / projection curves | SOFR OIS curves, initially single-curve framework | [A] |
| Initial upfront fee | Zero for the model; cannot infer from reporting dash | [A] |

Date update: the earlier 31 May 2023 suggestion is not a fact. We use 17/19 May in the baseline because a contemporaneous 17 May rate quote is available. Sensitivity tests should also try alternative dates in May 2023. Treat all assumed dates as replaceable once the actual confirmation is available.

## 5 Valuation checkpoints (independent fund observations)

| Report date | Fund-reported value (USD) | Source |
|---|---|---|
| 30 Jun 2023 | +275,600 | [S2] |
| 31 Dec 2025 | −4,989 | [S4] |
| 31 Mar 2026 | −33,019 | [S5] |
| 30 Jun 2026 | +117,354 | [S1] |

These are periodic fund-reported valuations, not a public series of executable swap transaction prices. Their changes are NOT equal to full-period trading P&L because coupon settlements, accrual, curve moves and financial-reporting presentation matter. Match valuation cut-off, accrued interest and clean/dirty value conventions before calculating errors.

## 6 Proposed research workflow

- **Historical data** Acquire timestamped USD SOFR OIS quotes (multiple maturities) plus published daily SOFR fixings. The overnight SOFR series alone is not enough for an IRS yield curve.
- **In-sample calibration** Use only curve data available up to 16 May 2023 for initial risk-factor estimation (target: ~500 trading observations, conditional on availability). Bootstrap/fit zero-discount curves using dated OIS quotes.
- **Out-of-sample testing** From the model inception in May 2023 through June 2026, value both legs each business day; calculate one-day hypothetical P&L including net cash-flow events. Predict 99% 1-day historical-simulation VaR and 97.5% ES using a rolling window based only on earlier observations.
- **Validation** Count VaR exceptions; compare static vs rolling-window methods. Separately compare model fair values with [S1], [S2], [S4] and [S5] on reported valuation dates, with a clear error decomposition.

- **Risk presentation** Show the fund's pay-fixed/receive-floating position and its receive-fixed/pay-floating mirror. Discuss parallel shifts, key-rate sensitivity, SOFR basis to the fund's OBFR-linked funding cost and valuation limitations.

## 7 Open items / limitations [U]

The fund has not supplied the underlying swap confirmation in the cited reports. The exact execution/effective/maturity dates, cash-flow schedules, SOFR observation shift/lookback, business-day rules, day counts, negotiated counterparty, any past modifications and exact accounting valuation adjustments remain unknown. Some reported terms could reflect amendments. We should not tune assumptions to force a match with SEC-reported values (look-ahead / overfitting risk).

## 8 Evidence & source links

**[S1] SEC 2026 semiannual report, fund's investments, printed p.10; Note 3 printed p.21; Note 8 pp.24–25**

https://www.sec.gov/Archives/edgar/data/925683/000092568326000008/f45913d1.htm

**[S2] SEC 2023 semiannual report, 30 Jun 2023, DERIVATIVES / SWAPS, printed p.10**

https://www.sec.gov/Archives/edgar/data/925683/000168386323006056/f26058d1.htm

**[S3] PNC / Texas Society of CPAs May 2023 presentation, "Cash Flow Hedge Terms and Indicative Pricing"; pricing as of 17 May 2023 (indicative 5Y rate 3.350%)**

https://www.tx.cpa/docs/librariesprovider2/2023-sae-materials/michael-stewart---share-tbd---may-2023-seminar-ms-final.pdf?sfvrsn=5c8ca8b1_1

**[S4] SEC 2025 annual report, 31 Dec 2025, DERIVATIVES / SWAPS**

https://www.sec.gov/Archives/edgar/data/925683/000119312526076989/f44069d1.htm

**[S5] SEC 2026 Q1 portfolio holdings, 31 Mar 2026**

https://www.sec.gov/Archives/edgar/data/925683/000092568326000006/edgar.htm

**[S6] CME SOFR term-structure data definitions / historical availability description**

https://www.cmegroup.com/market-data/faq-sofr-third-party-data.html

**Group decision requested: agree on the provisional May 2023 modelling dates and the listed conventions; retain the evidence/status labels in the final report. Final work requires independent acquisition of historical multi-tenor SOFR OIS data.**